

CSE 2320 - Homework 6

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Total points: 115/100 (points past 100 are bonus) Topics: Memoization, Greedy, Dynamic Programming (Knapsack: unbounded, 0/1, fractional)

P1 (4 pts) Given this solution information, for the **unbounded** Knapsack problem below, recover the choices that gave the optimal answer for knapsack capacity 19. Show your work (highlight or circle cells).

Item | A | B | C | D |
weight | 3 | 4 | 7 | 8 |
value | the item values are hidden as they should not be used in recovering the solution.

picked | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 |
| | | | A | B | B | A | C | D | D | A | B | B | A | C | C | D | A | B | B |

Items picked for capacity 19: ... B, C, D

P2 (61 pts) Given the item types below, solve the following problems. Fill in the answer in the table and show your work below.

Item:	A	B	C	D
Weight:	3	4	6	7
Value:	4	7	10	12

	Unbounded Knapsack	0/1 Knapsack	0/1, Fractional Knapsack
Dynamic Programming	\$\$: <u>24</u> Items: <u>B, B, C</u>	\$\$: <u>22</u> Items: <u>D, C</u>	
Greedy	\$\$: <u>21</u> Items: <u>B, B, B</u>	\$\$: <u>23</u> Items: <u>B, D, A</u>	\$\$: <u>24</u> Items: <u>B, D, $\frac{1}{2}C$</u>

a) (20 pts) Solve the **unbounded** Knapsack problem. Recover the items in the solution and show how you did that (e.g. highlight or circle cells). Show your work as done in class.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
max															
Sol	0	0	0	4	7	7	10	12	14	14	17	19	21	22	24
picked	-	-	-	A	B	B	C	D	B	A	B	B	B	C	B
A,3,4				0 4	1 4	2 4	3 8	4 11	5 11	6 14	7 16	8 18	9 18	10 21	11 23
B,4,7					1 7	2 7	3 7	4 10	5 14	6 14	7 17	8 19	9 21	10 21	11 24
C,6,10							0 10	1 10	2 10	3 14	4 17	5 17	6 20	7 20	8 24
D,7,12								0 12	1 12	2 12	3 12	4 19	5 19	6 22	7 24

Picked: B, B, C Value: 7+7+10 = 24

b) (20 pts) Solve the 0/1 Knapsack problem below (15pts). Use a star to show if the current item was used or not in the solution (8pts). Recover the items in the solution and show how you did that (e.g. highlight or circle cells) (7 pts). Show your work as done in class.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Sol															
no item	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A,3,4	0	0	0	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*
B,4,7	0	0	0	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*	0 4*
C,6,10	0	0	0	0	0	0	0 10*	0 10*	0 10*	0 10*	0 10*	0 10*	0 10*	0 10*	0 10*
D,7,12	0	0	0	0	0	0	0	0 12*	0 12*	0 12*	0 12*	0 12*	0 12*	0 12*	0 12*

Picked: D, C Value: 12+10 = 22

c) (8 pts) What items will a Greedy algorithm based on the ratio, choose for an **unbounded** Knapsack problem of size 14? Show your work.

A B C D
 4 7 10 12
 3 4 6 7
 1.33 1.75 1.66 1.71
 Picked: B, D, C, A

Picked: B B B
 4-4 10-4 6-4
 10 6 2
 Remaining: 8
 value: 7 7 7

total value: 21
 Picked: B, B, B

d) (8 pts) What items will a Greedy algorithm based on the ratio, choose for a 0/1 Knapsack problem of size 14? Show your work.

$A = 1.33$	
$B = 1.75$	
$C = 1.66$	
$D = 1.71$	

sorted: B, D, C, A	
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picked:	B	D	A
Remaining:	14-4	10-7	3-3
	10	3	0
Value:	7	12	4

total value: 23
 picked: B, D, A

e) (5 pts) What items will a Greedy algorithm based on the ratio, choose for a 0/1 Fractional Knapsack problem of size 14? Assume you have only one of each item. Show your work.

$A = 1.33$	
$B = 1.75$	
$C = 1.66$	
$D = 1.71$	

sorted: B, D, C, A	
--------------------	--

picked:	B	D	C
Remaining:	14-4	10-7	3-3
	10	3	0
Value:	7	12	$\frac{3}{6} \cdot 10$

total value: 24
 picked: B, D, $\frac{1}{2}C$

P3 (50 pts) Consider this recursive function:

```

int foo(int N){
    if (N <= 1) return 5;
    int res1 = 3*foo(N/2);
    int res2 = foo(N-1);
    if (res1 >= res2)
        return res1;
    else
        return res2;
}
    
```

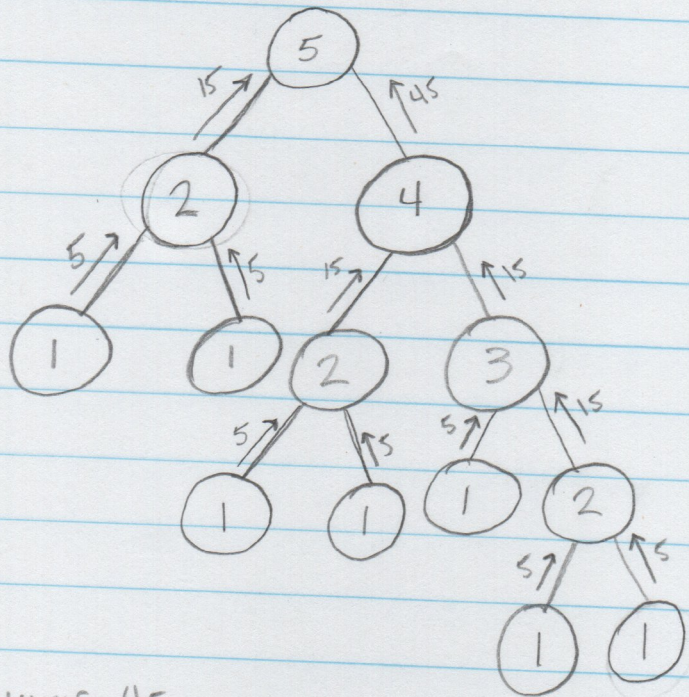
a) (6 points) Write the recurrence formula for the TIME COMPLEXITY of this function, including the base cases for $N \geq 0$. You do NOT need to solve the recurrence. Remember not to confuse the time complexity of the function with what the function calculates.

b) (8 points) Draw the tree that shows the function calls performed in order to compute $foo(5)$ (the root will be $foo(5)$ and it will have a child for each recursive call.) Also show what each call returns by using an arrow pointing back from the child to the parent.

P3

- a) if $N \leq 1$: $T(N) = C = 5$
if N is odd: $T(N) = T(N/2) + C$
if N is even: $T(N) = T(N-1) + C$

b)



$foo(5)$ returns 45